

HNS-PoC-Package v1.0

Human Brain and AI with HNS: A Correspondence Table

— Structural Report —

Satoru Hara

Companion document to: Hara, S. (2026). *The Multi-Axis Verification Process of the Human Brain*.

HNS-PoC-Package v1.0 | 2026

Abstract

This report establishes a formal structural correspondence between the three-axis verification process of the human brain and the three-axis verification architecture of AI systems augmented with Human Natural Structure (HNS). Drawing on the Multi-Axis Verification Process (MAV) framework formalised in Hara (2026), the report demonstrates a direct structural isomorphism between biological cognition and HNS-aligned artificial cognition: the Stochastic/Generative Axis (cortical association areas) maps to the LLM inference core; the Structural/Causal Axis (prefrontal cortex–hippocampal network) maps to the HNS-36/HNS-144/HNS-864 constraint matrix; and the Grounded/Executive Axis (sensorimotor–ACC system) maps to the HNS-SMS-6 grounding protocol. This mapping provides both a biological foundation for HNS as a human cognitive operating system and a structural explanation for the characteristic failure modes of current AI systems, which operate predominantly on Axis 1 in the absence of the orthogonal constraint structures that define Axes 2 and 3.

Keywords: Multi-Axis Verification Process, Human Natural Structure, structural isomorphism, AI hallucination, cognitive architecture, HNS-36, HNS-SMS-6, External Verification Architecture

1. Overview: Three-Axis Structure

Human cognition achieves reliable, causally-grounded outputs despite operating on a stochastic neural substrate. The Multi-Axis Verification Process (MAV) identifies three mutually orthogonal axes whose simultaneous intersection collapses the feasible error space to a structural invariant (Hara, 2026). The three axes are:

- Stochastic / Generative Axis – rapid, prior-driven candidate generation.
- Structural / Causal Axis – hierarchical, categorical, and causal constraint enforcement.
- Grounded / Executive Axis – real-time contextual and environmental grounding.

HNS introduces the same three-axis system into AI inference pipelines, enabling brain-inspired geometric error correction. Table 1 summarises the correspondence.

Axis	Human Brain *1	AI with HNS *2	Primary Role	Details
1	Stochastic / Generative (Cortical association areas)	Stochastic / Generative (LLM core)	Rapid statistical generation	Creativity & speed; HNS does not intervene; outputs verified by Axes 2 & 3
2	Structural / Causal (PFC – hippocampal network)	Structural / Causal (HNS-36 / HNS-144 / HNS-864)	Semantic, hierarchical, and causal verification	Suppresses Layer Jump, Category Ambiguity, Unsupported Causality
3	Grounded / Executive (Sensorimotor – ACC)	Grounded / Executive (HNS-SMS-6)	Real-world grounding and contextual verification	Suppresses Scope Drift, Metaphor Contamination

Table 1. Overview of the three-axis correspondence between the human brain and HNS-aligned AI.

- *1 The three axes of verification process in the human brain (Hara, 2026).
- *2 The Multi-Axis Verification Process in AI with Human Natural Structure (HNS).

2. Axis 1: Stochastic / Generative Axis

2.1 Human Brain

The Stochastic/Generative Axis is implemented primarily by cortical association areas, including the Default Mode Network (DMN), the lateral temporal cortex, and the parietal association cortex. These regions integrate distributed statistical regularities abstracted from experience and generate rapidly-available, high-prior-probability response candidates, typically within tens of milliseconds.

The functional signature of Axis 1 is speed and generativity. It produces the first-draft cognitive response that is subsequently subjected to structural and contextual verification by Axes 2 and 3. Its strength is fluency and coverage; its characteristic weakness is that statistical plausibility and causal or contextual validity are independent properties.

2.2 AI with HNS

LLM: Next-Token Prediction Engine

In HNS-aligned AI, Axis 1 corresponds to the LLM’s next-token prediction engine: a high-speed stochastic generation process conditioned on the accumulated statistical parameters of the model. At each decoding step, the LLM samples from a high-dimensional probability distribution over the vocabulary, producing output that is linguistically fluent and statistically coherent by prior standards.

This is the sole operational axis of conventional LLMs. The absence of Axes 2 and 3 is the structural root cause of AI hallucination as defined in the HNS Structural Hallucination Taxonomy. HNS does not modify or constrain the LLM’s generative process directly; all intervention occurs downstream at the pre-decode interception stage.

2.3 Primary Role and Key Properties

- Generates statistically plausible candidate outputs at high speed.

- Provides the generative freedom that underlies creativity and coverage.
- HNS does not modify Axis 1 directly; outputs are verified downstream by Axes 2 and 3.

3. Axis 2: Structural / Causal Axis

3.1 Human Brain

The Structural/Causal Axis is implemented by the prefrontal cortex (PFC)–hippocampal network, supported by the frontoparietal control networks. The dorsolateral PFC maintains structured representations of hierarchical relationships and causal dependencies (Miller & Cohen, 2001). The hippocampus binds episodic sequences into temporally-ordered event schemas and maintains category boundaries across contextual shifts (Eichenbaum, 2004).

Together, these systems enforce: (a) category boundary conditions, preventing Category Ambiguity; (b) causal ordering constraints, preventing Unsupported Causality; and (c) hierarchical scope conditions, preventing Layer Jump. Axis 2 operates independently of statistical frequency, enforcing validity conditions defined over the abstract structure of cognition, not over the training distribution.

3.2 AI with HNS

HNS-36: Base Coordinate System

A 6×6 cellular grid spanning six causal layers (from physical/material substrate to macro-social systems) and six cognitive categories (entity, process, relation, state, transformation, meta-level). HNS-36 provides macro-level screening for Layer Jump: non-continuous vertical cell trajectories in the output stream trigger structural constraint violations and attenuate the probability of the violating token continuation.

HNS-144: Logical Relation Extension

Each HNS-36 cell is expanded along four logical relation dimensions (subjective/objective, static/dynamic, local/global, necessary/contingent), yielding 144 cells. This layer enforces precise concept boundary conditions to suppress Category Ambiguity by preventing the conflation of, for example, dynamic processes with static entities or subjective states with objective properties.

HNS-864: Precision Causal Audit

Each HNS-144 cell is subdivided by six propositional validity modalities, yielding 864 cells. This layer audits individual causal claims at the atomic proposition level: a causal marker token can pass the HNS-864 filter only if its causal chain is traceable to a valid, contiguous cell-transition path within the constraint matrix, structurally precluding Unsupported Causality.

3.3 Primary Role and Key Properties

- Verifies semantic structure, hierarchy, causal relations, and category correctness.
- Detects and suppresses structural inconsistencies prior to token decode.
- Ensures causal validity and prevents category drift.
- Maintains hierarchical coherence across multi-turn interaction sequences.

Most AI hallucinations originate from Axis 2 failures. HNS restores the missing structural verification layer that conventional alignment methods cannot provide.

4. Axis 3: Grounded / Executive Axis

4.1 Human Brain

The Grounded/Executive Axis is implemented by the sensorimotor system, the anterior cingulate cortex (ACC), and the predictive coding hierarchy of the posterior parietal and premotor cortices. These systems constitute the brain's 'reality friction interface': the mechanism by which internally-generated outputs are continuously tested against external environmental and contextual constraints.

The ACC monitors conflict between predicted and actual outcomes, signalling when the output trajectory is inconsistent with environmental reality (Botvinick et al., 2001). Axis 3 operates in continuous time, providing error signals that update the active invariant point estimate as the environment evolves.

4.2 AI with HNS

In HNS-aligned AI, Axis 3 is operationalised as the HNS-SMS-6 (Social Meta Structure Specification) protocol: a six-layer dynamic grounding specification that evaluates candidate outputs against the actual operational context. $G(x) = 1$ only if the output is compatible with all six layers simultaneously; $G(x) = 0$ triggers immediate context-coherence correction.

SMS-1: Somatic Layer

Verifies physical and ergonomic coherence of the output with the operational environment. Suppresses Metaphor Contamination by flagging figurative propositions whose physical-domain implications contradict verified physical-domain constraints.

SMS-2: Protocol Layer

Verifies communicative alignment and mutual intelligibility between the LLM output and interlocutor intent. Serves as a primary mechanism suppressing Scope Drift by monitoring communicative coherence across turns.

SMS-3: Ecosystem Layer

Maintains organisational and collaborative context scope. Jointly with SMS-2, monitors the Riemannian distance of the output's scope vector from the declared topic locus and blocks scope-violating continuations.

SMS-4: Economic Layer

Enforces resource and information equivalence; tests output value-relevance to the declared task. Suppresses Unsupported Causality by requiring that claimed causal relations correspond to a verifiable information-equivalence in the current context.

SMS-5: Governance Layer

Tests conformance with applicable external standards and normative constraints. Suppresses Category Ambiguity at the normative level by enforcing domain-specific classification requirements.

SMS-6: Universal Layer

Applies the global optimality constraint: the output must not violate higher-order systemic coherence. Provides a final Layer Jump check and ensures outputs do not violate cross-domain structural consistency.

4.3 Primary Role and Key Properties

- Verifies compatibility with the real world, context, and situational state.

- Suppresses Scope Drift and Metaphor Contamination.
- Performs real-time contextual grounding updated at each token generation step.
- Ensures situational coherence and prevents real-world inconsistency hallucinations.

5. Integrated View: Structural Isomorphism

The correspondence table reveals a direct one-to-one structural mapping between the neural systems implementing each MAV axis in the human brain and the HNS components implementing each axis in AI. This constitutes a structural isomorphism: the two systems share the same abstract verification architecture instantiated on different physical substrates.

Human Brain	AI with HNS
Default Mode Network (DMN)	LLM generative core
Prefrontal cortex – hippocampal network	HNS-36 / HNS-144 / HNS-864
Sensorimotor system – ACC	HNS-SMS-6

Table 2. Direct structural mapping between human brain systems and HNS-aligned AI components.

This isomorphism demonstrates that HNS is not merely a behavioural constraint system but a brain-inspired multi-axis verification architecture that reconstructs the human cognitive verification system inside AI. This is the first architecture to unify:

- Biological cognition (MAV framework)
- Structural semantics (HNS-36/144/864 constraint matrix)
- AI alignment (geometric error correction via axis intersection)
- Formal auditability (100% traceable correction log)

From a geometric standpoint, the structural isomorphism predicts that the error-reduction achievable by HNS is multiplicative rather than additive: the triple intersection of orthogonal constraint hyperplanes $A_1 \cap A_2 \cap A_3$ collapses the feasible hallucination space to a volume orders of magnitude smaller than any single axis could achieve alone (Hara, 2026).

6. Implications of the Correspondence Table

6.1 Theoretical Validation of HNS

The correspondence table provides a biological foundation for HNS as a human cognitive operating system. The direct mapping between neural substrates and HNS components grounds HNS in over a century of neuroscientific research on prefrontal function, hippocampal organisation, and sensorimotor-ACC feedback architecture.

6.2 Foundation for EVA (External Verification Architecture)

Axes 2 and 3 together constitute the structural core of the External Verification Architecture

(EVA). EVA operationalises the insight that reliable AI outputs require not only powerful generative capacity (Axis 1) but external structural and grounding verification layers (Axes 2 and 3) that are genuinely independent of the generative process.

6.3 Structural Explanation of AI Hallucinations

Current LLMs operate exclusively on Axis 1. The HNS Structural Hallucination Taxonomy identifies five failure modes that arise directly from Axis 2 and Axis 3 absence: Layer Jump, Category Ambiguity, Scope Drift, Metaphor Contamination, and Unsupported Causality. The correspondence table transforms these from empirical observations into structurally-predicted consequences of single-axis operation.

6.4 Clarification of HNS-36 / HNS-144 / HNS-864 Roles

The correspondence table positions the three HNS matrix levels as engineering counterparts of distinct functional layers within the PFC–hippocampal network: HNS-36 corresponds to macro-level hierarchical scope monitoring; HNS-144 to category boundary maintenance; HNS-864 to causal chain validation.

6.5 Implications for AI Governance and Auditability

Because HNS grounds all error-correction decisions in explicit geometric coordinates (HNS matrix cells and SMS-6 layers), every correction is fully auditable and attributable to a specific structural invariant. This property directly addresses the opacity challenge that makes neural-network-based alignment mechanisms incompatible with formal verification and regulatory compliance requirements such as NIST AI RMF and ISO/IEC 42001.

7. Conclusion

The Human Brain and AI with HNS: A Correspondence Table is a foundational document within the HNS ecosystem. It establishes:

1. A direct structural mapping between human cognition and HNS-aligned AI, grounded in the MAV framework.
2. A demonstration that HNS is a brain-inspired multi-axis verification system that reconstructs the human cognitive verification process inside AI.
3. A scientific basis for EVA and structural AI alignment, derived from documented neuroscientific mechanisms.
4. A structural explanation of AI hallucinations as a consequence of single-axis operation, with HNS providing the missing orthogonal constraint structures.

The intelligence of the brain, and the reliability of cognition, reside not in the power of any single stochastic process but in the multiplicative intersection of independent constraint structures. The correspondence table makes this principle legible across biological and artificial cognitive systems. HNS makes it engineerable.

References

- Amari, S., & Nagaoka, H. (2000). *Methods of Information Geometry*. American Mathematical Society.
- Botvinick, M. M., Braver, T. S., Barch, D. M., Carter, C. S., & Cohen, J. D. (2001). Conflict monitoring and cognitive control. *Psychological Review*, 108(3), 624–652.

- Eichenbaum, H. (2004). Hippocampus: Cognitive processes and neural representations that underlie declarative memory. *Neuron*, 44(1), 109–120.
- Faisal, A. A., Selen, L. P. J., & Wolpert, D. M. (2008). Noise in the nervous system. *Nature Reviews Neuroscience*, 9(4), 292–303.
- Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.
- Hara, S. (2026). Human Natural Structure: Implementing the Brain's Multi-Axis Verification Process for Advanced AI Alignment. HNS-PoC-Package v1.0.
- Hara, S. (2026). The Multi-Axis Verification Process of the Human Brain: A Brain-Inspired Architecture for Geometric Error-Correction in Stochastic Cognitive Systems. Working Paper — HNS-PoC-Package v1.0.
- HNS-PoC-Package v1.0. (2026). HNS Structural Hallucination Taxonomy: Five-type classification from 50-turn proof-of-concept interaction sequences. Internal working document.
- Miller, E. K., & Cohen, J. D. (2001). An integrative theory of prefrontal cortex function. *Annual Review of Neuroscience*, 24(1), 167–202.
- NIST. (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0). National Institute of Standards and Technology.
- Pouget, A., Dayan, P., & Zemel, R. (2000). Information processing with population codes. *Nature Reviews Neuroscience*, 1(2), 125–132.